

Ahmad Chalhoub

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Summary

Perception Engineer leading 3D sensor fusion and real-time deep learning for Mitsubishi Electric's [HubPilot project](#). Pursuing an MEng in CSE at the University of Michigan, Ann Arbor.

Experience

Automotive Perception Engineer III

Mitsubishi Electric Automotive America • Northville, MI

09/2024 - Present

- Lead end-to-end development, integration, and testing of HubPilot's perception system (HubDrive), exceeding accuracy requirements.
- Research, prototype, and develop advanced 3D perception models, emphasizing sensor fusion approaches.
- Solely designed, developed, deployed, and tested HubPilot's YardPass feature.
- Optimize deep learning models and perception software for NVIDIA embedded SoCs.
- Collaborate with cross-functional teams in Japan and research scientists at MERL on development efforts.

Automotive Perception Engineer II

Mitsubishi Electric Automotive America • Northville, MI

01/2022 - 09/2024

- Spearheaded the design and development of HubPilot's HubDrive system from concept to initial deployment, ensuring robust architecture and on-schedule delivery.
- Orchestrated data collection and labeling for multiple RGB camera deep learning models.
- Translated overall system and customer requirements into multi-level perception specifications.

Machine Learning Research Intern

Mitsubishi Electric Automotive America • Northville, MI

05/2021 - 12/2021

- Led MEAA's first in-house High-Performance Compute (HPC) machine build for on-site data processing.
- Developed a deep learning vision solution for Qualcomm's SoC (showcased at CES) using the Neural Processing SDK.

Research Experience

University of Michigan, Ann Arbor (01/2025 - Present)

- Using VLMs for end-to-end autonomous driving; [RobustNet group](#) (advised by Prof. Morley Mao)

Education

MEng - Computer Science & Engineering

University of Michigan • Ann Arbor, MI

12/2027

BEng - Robotics and Mechatronic Systems

University of Detroit Mercy • Detroit, MI

12/2021

Skills

3D Computer Vision, Multimodal Sensor Fusion, Python & C++, PyTorch, ROS2, Git, Linux, Deep Learning, Leadership